

Education

City University of Hong Kong	Hong Kong SAR
B. E. in Computer and Data Engineering, Department of Electrical Engineering	September 2019 - June 2025
Coursework: Cloud Computing Systems, Database Systems, Big Data, and Human-Computer Interaction	
Vanderbilt University	Nashville, TN, USA
Study abroad coursework in the School of Engineering	January 2024 - May 2024
Coursework: Applied Machine Learning, Artificial Intelligence, FPGA Design, and Programming Languages	

Technical Skills

Programming Languages	Python, JavaScript/TypeScript, C/C++, Rust
Frameworks and Tools	Frontend/Mobile: React, Next.js, React Native, Unit Integration, E2E Testing Backend and Database: NestJS, FastAPI, PostgreSQL DevOps & Infrastructure: Git, CI/CD (GitHub Actions), AWS, Docker

Professional Experience

URJOB	Seoul, South Korea
Frontend Engineering Intern	March 2026 - April 2026

- Monorepo Migration: React + React Router Legacy to TanStack Start
- Separated user app and admin dashboard into a pnpm workspace monorepo, sharing runtime and design tokens
 - Replaced localStorage-backed JWT with HttpOnly + SameSite cookies to mitigate XSS and CSRF attack vectors
 - Integrated TanStack Query as the session source of truth and defined the access-token refresh policy
- Developer Experience Improvements
- Established project conventions and an agentic coding harness, enforcing formatting, lint, and commit standards via Lefthook to onboard developers and AI agents consistently
 - Generated TypeScript types from the OpenAPI spec to eliminate contract drift at compile time
 - Unified query keys and mutations via a Query/Mutation Factory for predictable cache invalidation across the project
- Deployment Pipeline & Infrastructure
- Designed a reusable workflow_call-based GitHub Actions architecture separating CI (lint/test), Build, and Deploy across develop, preview, and production environments
 - Moved Dockerfile-based EC2 builds into GitHub Actions, shipping only the .output artifact to minimize runtime image size and remove instance build load

MotoNerv Limited	Hong Kong SAR
Hardware (Embedded Systems) Engineering Intern	June 2023 - August 2023
- Developed a prototype of an MCU-based (ESP32) dashcam for streaming video and collecting sensor data - Preprocessed on-car sensor and location data tailored for AI video analytics in the AWS cloud environment - Implemented data pipeline to stream sensor and video data directly to AWS S3 buckets for further processing via cellular network connection	

Projects

RunFit	
Running crew matching platform with location-based filtering and pace-based discovery	
- Designed a domain-based REST API structure with typed DTO request/response bodies, standardizing the frontend-backend interface contract across a 4-person team through iterative requirement changes - Architected a Next.js server proxy layer to centralize API request/response control, preventing backend endpoint exposure and closing JWT token-handling vulnerabilities across App Router server/client component boundaries - Implemented role-based access control (RBAC) with httpOnly cookie-backed JWT, addressing secure session state across server and client components in Next.js App Router - Integrated TanStack Query with custom factory patterns for queries/mutations, establishing a structured error handling strategy across all data fetching layers	

Squash Motion Tracking (Final Year Project)



Participated in Hong Kong Sport Institute (HKSI) initiated computer vision project for athlete performance analysis

- Architected a full-stack computer-vision pipeline (FastAPI + React) for player tracking and pose detection from match video, enabling data-driven coaching
- Orchestrated SAM2 and YOLO model inference behind a REST API for video processing
- Reduced VRAM during transformer-based segmentation via frame sampling on forward/backward propagation
- Containerized the system with Docker / Docker Compose for reproducible deployment with NVIDIA GPU acceleration

Leadership Activities & Other Experiences

Frontend Short-Term Intensive Course - codeit

November 2025 - January 2026

- Completed 10-week intensive covering Jest/React Testing Library, CI/CD pipelines, and Next.js architecture patterns
- Established team workflows by defining code conventions, PR templates, and review processes for a 4-person team
- Led architecture decisions for API structure, auth flows, and state management in a production-grade running crew platform

Data Science Coaching Study: 2024 - boostcourse

July - August 2024

- Completed 4-week program on Python data analysis covering pandas, numpy, and visualization libs (matplotlib, seaborn, plotly)
- Practiced EDA techniques including correlation analysis, pivot tables, and statistical visualization on real-world KOSIS datasets

Student Resident (Senior Resident Tutor)

July 2024 - July 2025

- Provided academic mentoring and pastoral support to resident students; coordinated hall community events and welfare programs

Student Resident Cabinet (General Officer)

October 2022 - May 2023

- Contributed to student welfare initiatives and community programs as a student government officer

Military service

August 2020 - February 2022

- Completed mandatory Republic of Korea military service

Private Tutoring

- Taught International Baccalaureate (IB) Diploma Mathematics and Electrical Engineering to secondary and university-level students